

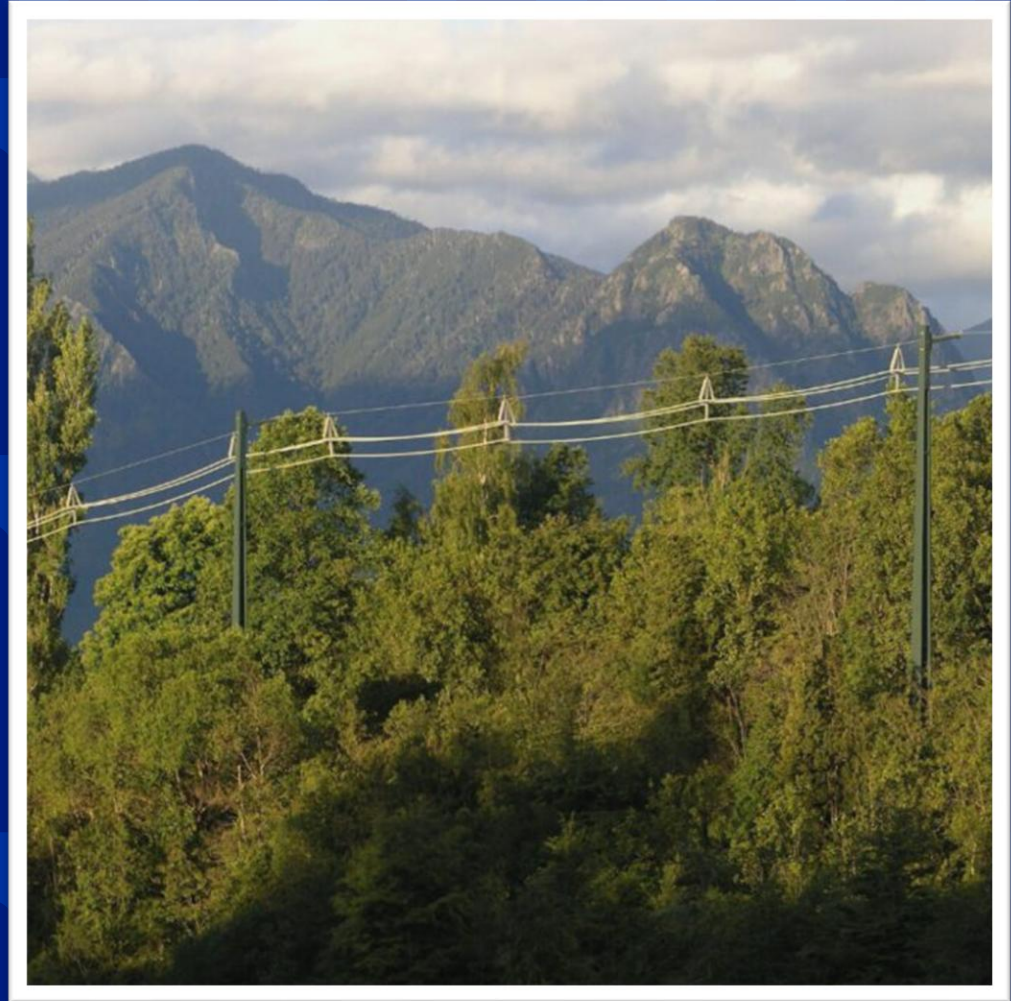
RESEARCH & DESIGN WITH MARMON UTILITY

Matthew Cromer | Mechanical Engineering Intern



Marmon
Electrical

A Berkshire Hathaway Company



About Me



I am a Mechanical Engineering Student at the University of Connecticut going into my senior year.

I am pursuing a double concentration in Dynamic Systems and Control (Robotics) as well as in Design and Manufacturing.



I am a Mechanical Engineering Intern at Marmon Utility in Amherst, NH, mostly working with the Molded team.



Fun Personal Fact

I am a big fan of anything that is outdoors. Below is a picture of me at Yosemite in front of Half Dome



AGENDA

1. Project
2. Observations & Areas Of Opportunities
3. Recommended Solutions
4. Takeaways
5. Q & A



Project Context



Assignment

Previous testing setup and results

Results Compiled

Test	# of Clamps	Bolt Torque	Test Rate	Max Load
1	1	Hand Tight	.5 in/min	111 lbs.
2	1	30 ft-lb	.5 in/min	129 lbs.
3	2	40 ft-lb	.5 in/min	211 lbs.
4	2	40 ft-lb	.5 in/min	219 lbs.
5	2	40 ft-lb	2 in/min	253 lbs.
6	2	40 ft-lb	4 in/ min	258 lbs.
7	3	40 ft-lb	4 in/min	344 lbs.
8	3	40 ft-lb	.5 in/min	278 lbs.
9	3	40 ft-lb	.5 in/min	308 lbs.

Goal is 1400 lbs.



Research & Development

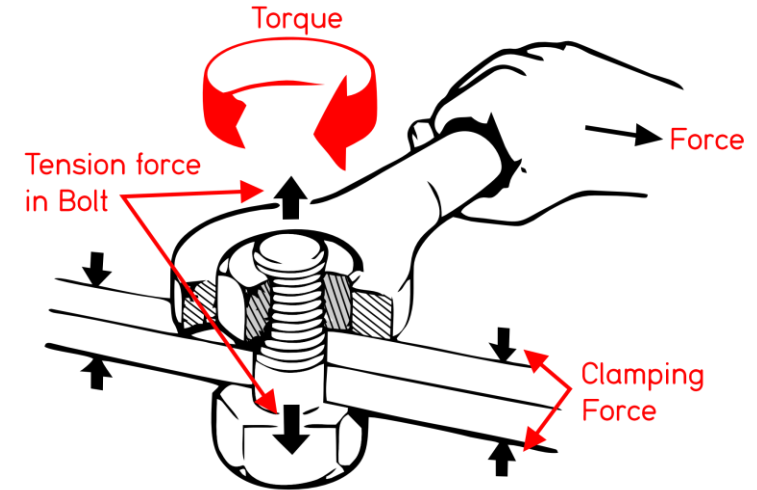
The idea to increase clamping force was to implement inserts, this branched into two separate ideas.

1. Design a new curved insert
2. Implement an already established insert

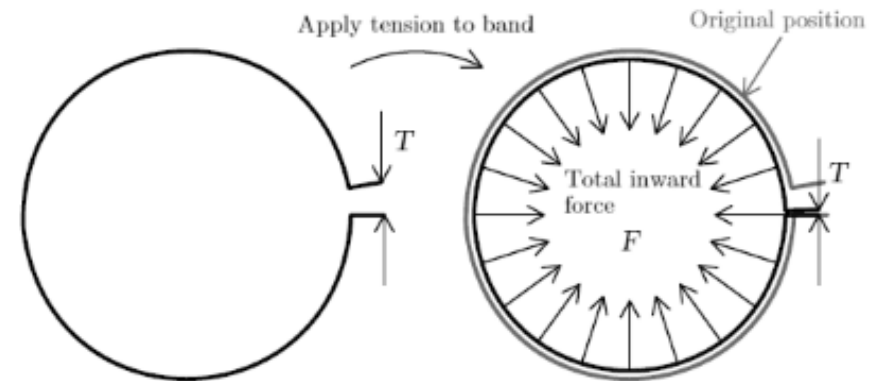
Vice Top Insulators with Ceramic Inserts



How Torque increases Clamping Force

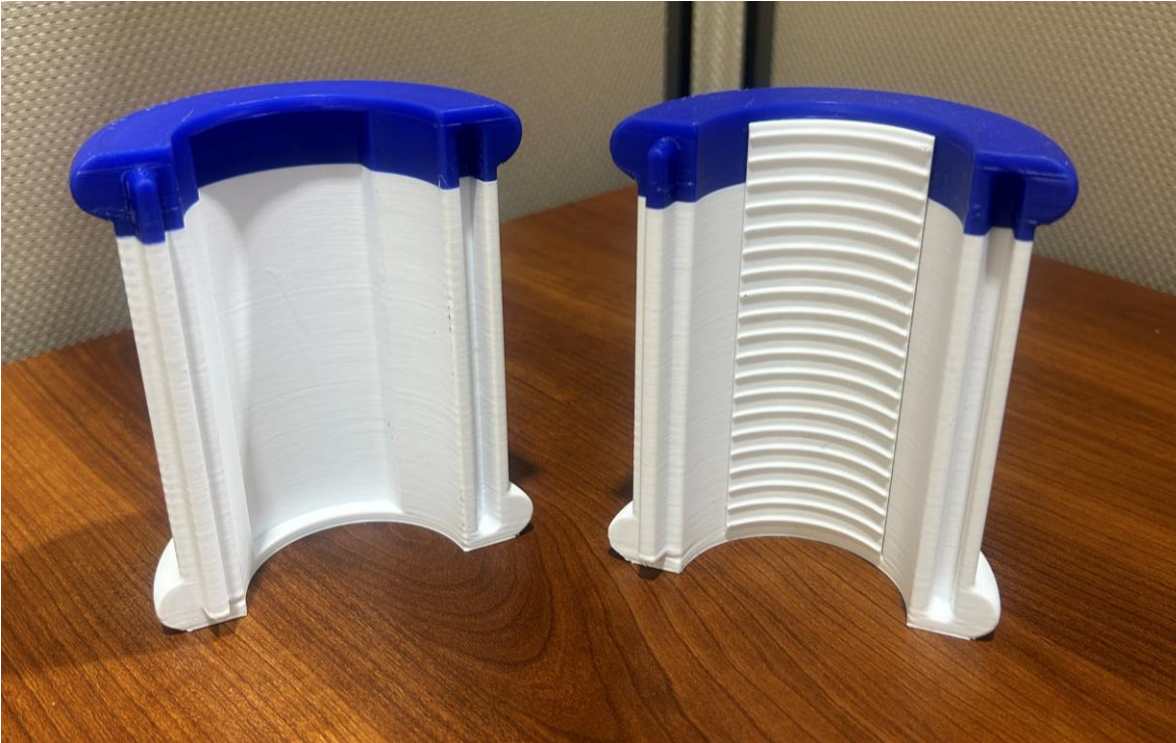


Increasing Clamping Force



Curved Insert: First Iteration

Prototype

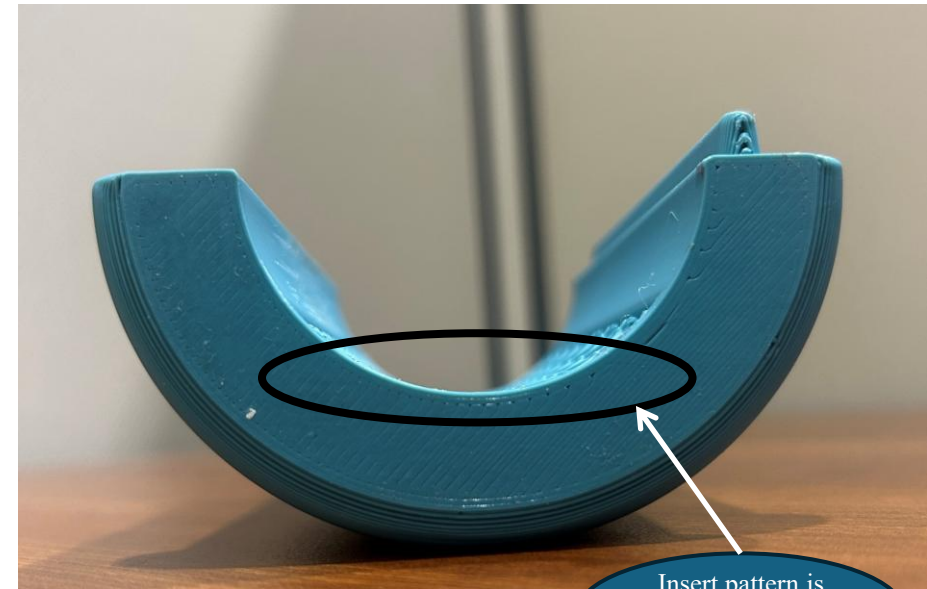


Issue



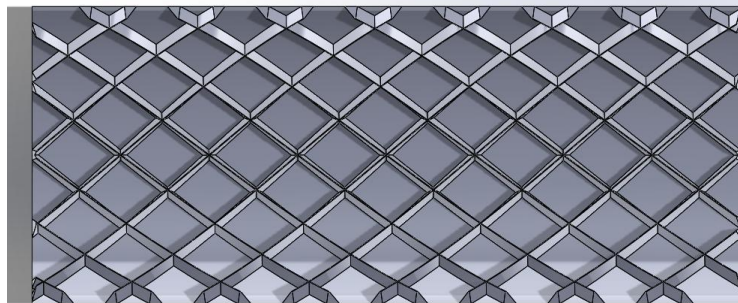
Curved Insert: Second Iteration

Prototype



Insert pattern is not prominent enough

SolidWorks screenshot of new pattern

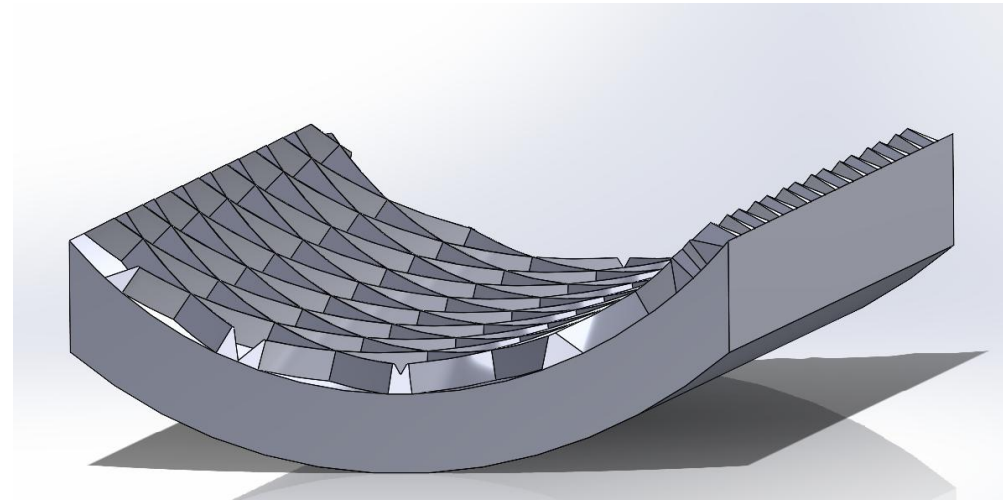


Curved Insert: Final Iteration

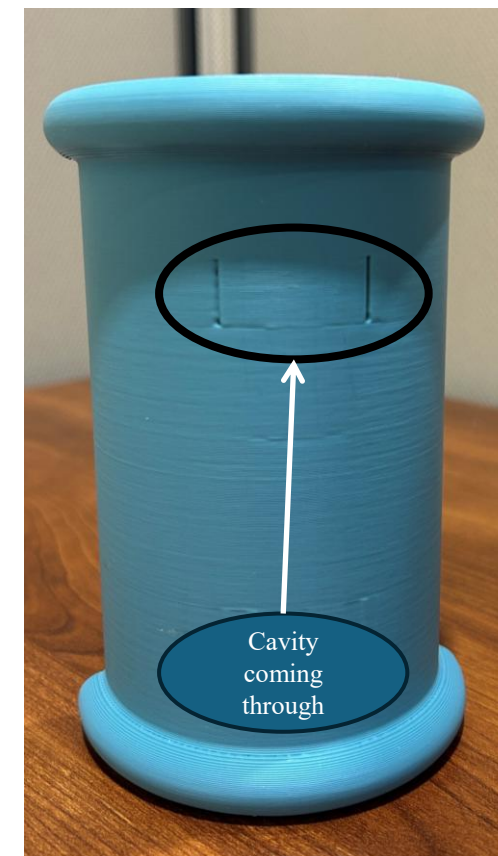
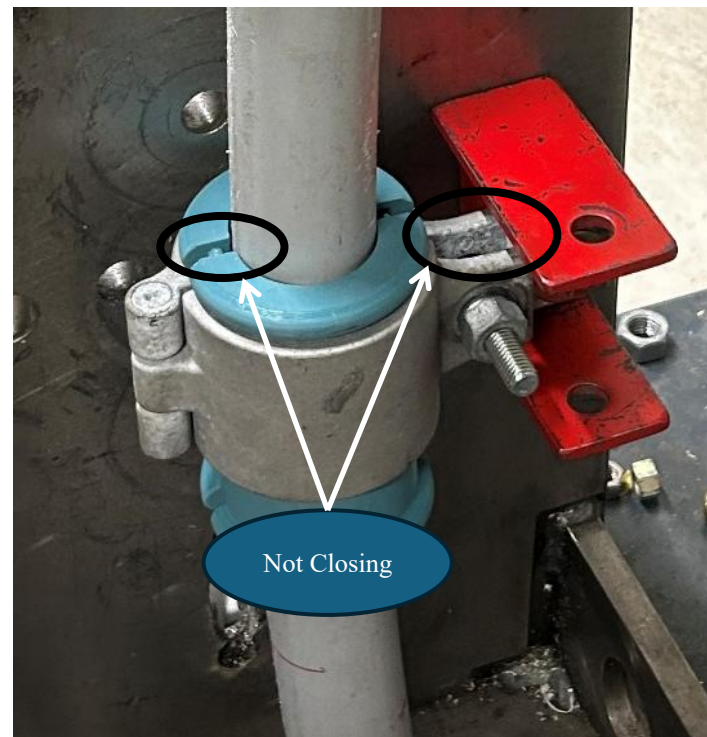
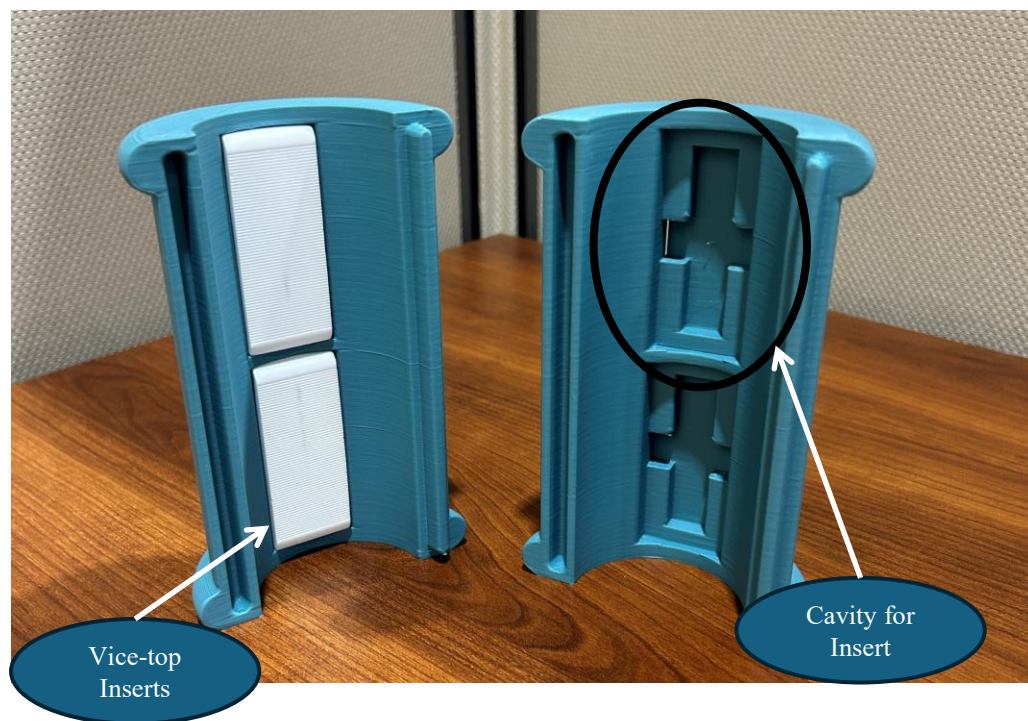
Final Design



Final Insert Design

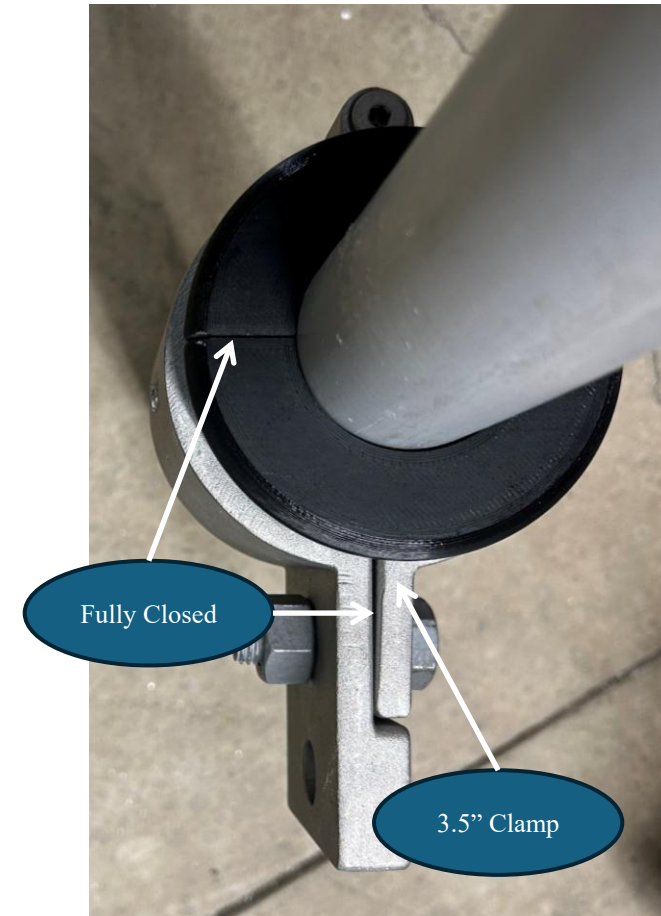


Ceramic Insert: First Iteration



*Want 1/4 inch of material between the back of the insert and the edge of the wall

Ceramic Insert: Final Iteration



Overall Process

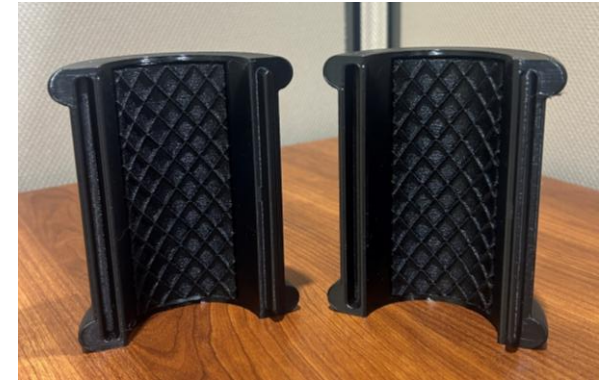
Parallel ridge insert



Crisscross insert V1



Crisscross insert V2



Ceramic inserts V1



Ceramic inserts V2



Results

New testing results

Ceramic Insert			
Test	Test Rate (in/min)	Bolt Torque (ft-lb)	Load (lb)
1	0.5	20	1400
2	0.5	30	1400
3	0.5	40	1400
4	0.5	50	1886

3D Printed Insert			
Test	Test Rate (in/min)	Bolt Torque (ft-lb)	Load (lb)
1	0.5	20	572.66
2	0.5	30	624.48
3	0.5	30	626.10
4	0.5	40	705.19
5	0.5	40	708.36
6	0.5	50	737.39

Old testing results

Results Compiled

Test	# of Clamps	Bolt Torque	Test Rate	Max Load
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Ceramic Inserts

129 pounds → 1886 pounds ≈ 1300% increase

3D Printed Inserts

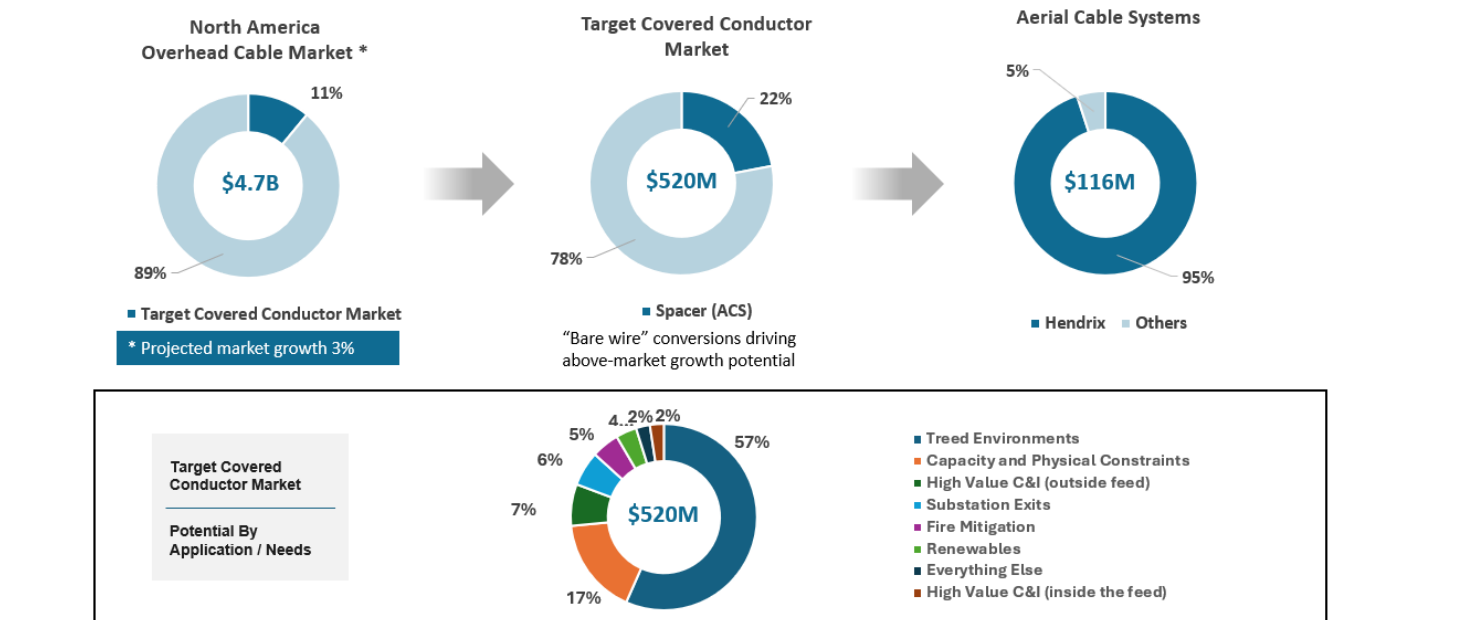
129 pounds → 737.39 pounds ≈ 470% increase



Observations and Areas of Opportunities

- This boot that provides more clamping power can be utilized in more ways than just the specific 115 kV arrangement. In any situation where it is imperative for the conductor to stay in secure contact with the insulator, this boot can be used. I believe there could be applications for the higher voltage systems such as the 145 kV.
- Because this design opens the door to a much larger market, our regional sales and product manager predicts a 5-9% growth in the potential market over the next 5 years.

Addressable Market – Overhead Solutions



Resolution and Recommendations

- As the full implementation of a new HDPE boot takes much longer than my time here, what I have done is shown proof of concept.
- As the typical vice-top inserts achieved the targeted value, that is the design that should be used for now
- It would be my recommendation that modifications be made to the current boot mold to allow space for an insert, or a new mold entirely is used, then more tests both in a lab environment as well as a real-world environment to continue to show the application for this new design.
- In the meantime, development of a curved ceramic insert with a crosshatch pattern would likely be the most effective solution.



Takeaways & Acknowledgements



Special Thanks

Special Thanks to

- Sonny Mercedat
- Josh Amman
- Richard Heckbert

Impact On Me

During my time here at Marmon gave me everything I was hoping for. I was able to take the knowledge I learned in the classroom and apply it to specific problems. I also learned more about specific nuances in engineering and really enjoyed working alongside other engineers.

Questions?